Final Exam Calculus 2019 2020 Sem 3

CALCULUS - foundation (Universiti Teknologi Malaysia)

QUESTION 1 (12 MARKS)

a) Function f is given by

$$f(x) = \begin{cases} (A-x)^2, & x < 4 \\ B + \sqrt{x}, & 4 \leq x < 9 \\ x - 4, & x \geq 9 \end{cases}$$

(i) Find the values of A and B if f is continuous everywhere.

(4 marks)

(ii) If $A = 2$ and $B = 3$, show that f is not continuous at $x = 4$.

(3 marks)

b) Without using L'Hopital's rule, evaluate

$$\lim_{x \to 2} \frac{\sqrt{6-x} - 2}{3 - \sqrt{x+7}}.$$

(5 marks)

QUESTION 2 (15 MARKS)

a) If $y = e^{\cos^4(\pi x)}$, find $\dfrac{dy}{dx}$.

(4 marks)

b) Given $y = e^{2x} \ln(x+2)$. Show that

$$(x+2)^2 \left(\frac{d^2y}{dx^2} - 2\frac{dy}{dx} \right) = e^{2x}(2x+3).$$

(6 marks)

c) Given

$$y^{\sin x} = x^{\frac{3}{2}}.$$

By taking natural logarithm on both sides, find $\dfrac{dy}{dx}$ in terms of x and y.

(5 marks)

QUESTION 3 (17 MARKS)

a) The cost in Malaysia Ringgit of employing n number of people in a restaurant is modelled by

$$C = n^3 - 27n^2 + 195n + 800, \quad \text{where } n \geq 4.$$

Calculate the minimum cost and the number of people who should be employed in order to minimize the cost.

(5 marks)

b) The function f is defined by $f(x) = x^5 - 15x^3$.

(i) Find the points when f crosses the x-axis correct to two decimal places.

(2 marks)

(ii) Show that f has an inflection point at $(0, 0)$. Then determine the minimum and maximum points.

(7 marks)

(iii) Hence by using the results from (i) and (ii), sketch the graph of f. Label all the important points.

(3 marks)

QUESTION 4 (16 MARKS)

a) Use the substitution $x = \sin\theta$ to evaluate

$$\int_{0.5}^{1} x^3\sqrt{1 - x^2}\, dx.$$

(8 marks)

b) Show that

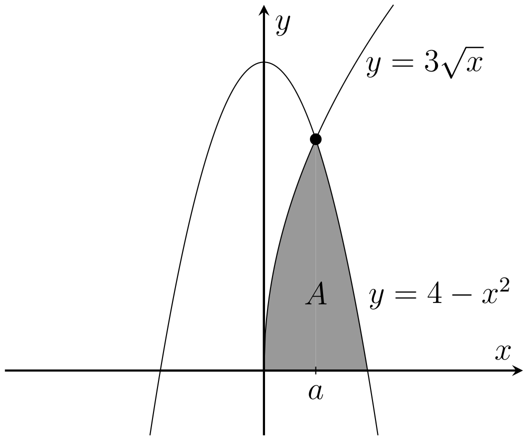
$$\frac{x^4 - x^3 - 4x^2 + 2x + 2}{x(x+1)^2} = x - 3 + \frac{A}{x} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}.$$

Find the values of A, B, and C. Hence, evaluate $\displaystyle\int \frac{x^4 - x^3 - 4x^2 + 2x + 2}{x(x+1)^2}\, dx$.

(8 marks)

QUESTION 5 (12 MARKS)

In the figure below, shaded region A is bounded by the curves $y = 4 - x^2$, $y = 3\sqrt{x}$, and x-axis. The curves intersect at a point where a is the x-coordinate.



(i) If a is an integer, find the value of a.

(2 marks)

(ii) Find the area of region A.

(4 marks)

(iii) Find the volume generated when the region A is rotated around x-axis completely.

(6 marks)

QUESTION 6 (11 MARKS)

a) A bacterial population is growing according to initial value problem

$$\frac{dP}{dt} = 0.2P, \qquad P(0) = 10\,000,$$

where P is the number of bacteria and t is time in hours. Solve the initial value problem. Hence find the number of bacteria after 5 hours.

(5 marks)

b) Given an ordinary differential equation

$$2x\frac{dy}{dx} + y = 2x^{\frac{3}{2}} \cos x.$$

Show that the integrating factor ρ, is given by $\rho = \sqrt{x}$. Hence solve the ordinary differential equation.

(6 marks)

QUESTION 7 (17 MARKS)

a) Given $f(x) = e^x + x^2 - 8x$.

(i) By using Intermediate Value Theorem, show that there are roots located in the interval $[0, 1]$ and $[2, 3]$.

(3 marks)

(ii) By taking an iterative

$$x_{i+1} = \frac{e^{x_i} + x_i^2}{8}, \qquad i = 0, 1, 2, \dots ,$$

find the root correct to four decimal places. Use $x_0 = 0.1$.

(4 marks)

(iii) Use Newton-Rhapson method to find the second root correct to four decimal places with $x_0 = 2$.

(5 marks)

b) By using Simpson's rule, compute the integral $\int_0^4 \ln(1+9x+x^3)\, dx$ with eight strips.

(5 marks)

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
• $1 + \tan^2 x = \sec^2 x$	• $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
• $\cot^2 x + 1 = \operatorname{cosec}^2 x$	• $\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
• $\sin 2x = 2 \sin x \cos x$	• $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$
• $\cos 2x = \cos^2 x - \sin^2 x$	• $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$
$= 2\cos^2 x - 1$	• $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
$= 1 - 2\sin^2 x$	
• $\tan 2x = \dfrac{2 \tan x}{1 - \tan^2 x}$	

Logarithm
$a^x = e^{x \ln a}$
$\log_a x = \dfrac{\log_b x}{\log_b a}$

Differentiations	**Integrations**
$\dfrac{d}{dx}[k] = 0,\ k$ constant	$\int k\,dx = kx + C,\ k$ constant
$\dfrac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \dfrac{x^{n+1}}{n+1} + C,\ n \neq -1$
$\dfrac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\dfrac{d}{dx}[\ln\|x\|] = \dfrac{1}{x}$	$\int \dfrac{dx}{x} = \ln\|x\| + C$
$\dfrac{d}{dx}[\cos x] = -\sin x$	$\int \sin x\,dx = -\cos x + C$
$\dfrac{d}{dx}[\sin x] = \cos x$	$\int \cos x\,dx = \sin x + C$
$\dfrac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x\,dx = \tan x + C$
$\dfrac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x\,dx = -\cot x + C$
$\dfrac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x\,dx = \sec x + C$
$\dfrac{d}{dx}[\operatorname{cosec}\ x] = -\operatorname{cosec}\ x \cot x$	$\int \operatorname{cosec}\ x \cot x\,dx = -\operatorname{cosec}\ x + C$

Parametric Differentiation
$\dfrac{dy}{dx} = \dfrac{dy}{dt} \times \dfrac{dt}{dx}, \qquad \dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \times \dfrac{dt}{dx}$
Integration by Parts
$\displaystyle\int u\ dv = uv - \int v\ du$
Fixed Point Formula
$x_{i+1} = g(x_i), \quad i = 0, 1, 2, \ldots$
Newton-Raphson Formula
$x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}, \quad i = 0, 1, 2, \ldots$
Trapezoidal rule
$\displaystyle\int_{x_0}^{x_N} f(x)\,dx = \frac{h}{2}\Big[f_0 + f_N + 2(f_1 + f_2 + \cdots + f_{N-1})\Big],\ h = \frac{x_N - x_0}{N}$
Simpson's rule
$\displaystyle\int_{x_0}^{x_N} f(x)\,dx = \frac{h}{3}\Big[f_0 + f_N + 4(f_1 + f_3 + \cdots + f_{N-1}) + 2(f_2 + f_4 + \cdots + f_{N-2})\Big],\ h = \frac{x_N - x_0}{N}$